MEDICAL CORE · VERIFIED

rTMS Coil Navigation Software (NeuroPilot)

Turn optical tracking and 3D Slicer into a license-gated coil-navigation product that gives operators position and angle guidance they can rely on.

EVIDENCE STATE **Verified · Ongoing**	PERIOD **2024.07 - present**
PORTFOLIO TIER **Medical Core**	PRIMARY CAPABILITY **Medical Navigation & Visualization**

Designed the 3D Slicer workflow from procedure setup to real-time coil navigation, the landmark-and-ICP registration engine, tracker, tablet, and robot integration, and a license-gated product structure.

Problem and system boundary

01 / NAVIGATION UI WORKFLOW

Navigation UI workflow

Six Home-screen menus gate the run from DICOM load through reconstruction, registration, and targets to the live session, with crosshair aiming.

PROBLEM

Medical images, a tracker, a tablet, and a robot each sit in their own coordinate frame and had to become one verifiable flow.

Designed the 3D Slicer workflow from procedure setup to real-time coil navigation, the landmark-and-ICP registration engine, tracker, tablet, and robot integration, and a license-gated product structure.

PUBLIC BOUNDARY

No clinical efficacy, quantitative accuracy, or regulatory status is claimed, and patient data stays out of this case.

The segmentation and landmark models were supplied externally and integrated behind an adapter boundary. I own that boundary, the post-processing, the license gating, and the UI integration.

EVIDENCE STATE

Verified · Ongoing / NeuroPilot navigation module: tracker, patient-marker and coil-marker status, distance and tilt guidance to the target, and the aiming guide (project path hidden).

Owned decisions and implementation

MY ROLE

Led the 3D Slicer procedure-setup and navigation workflow, the registration engine combining landmarks and ICP with its coordinate-transform chain, the optical-tracker SDK migration with the tablet protocol and robot integration, and the product structure covering license verification and optional-feature gating.

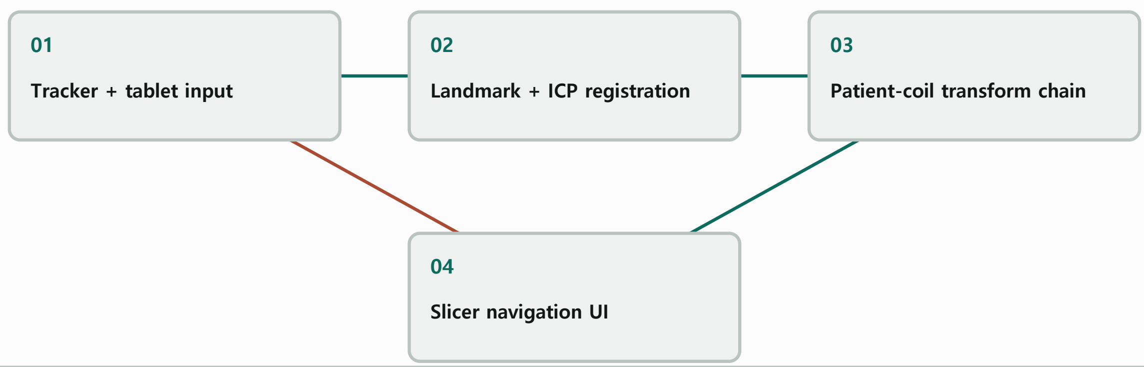
02 / COORDINATE AND REGISTRATION ENGINE

Coordinate and registration engine

Image, patient, tracker, and coil transforms stay explicit; ICP offers three modes.

SYSTEM RELATIONSHIP

Coordinate loop from input to aiming guidance



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

3D Slicer / VTK / PyQt5 / Python / PyTorch / Optical tracking / TCP/IP binary protocol / License gating

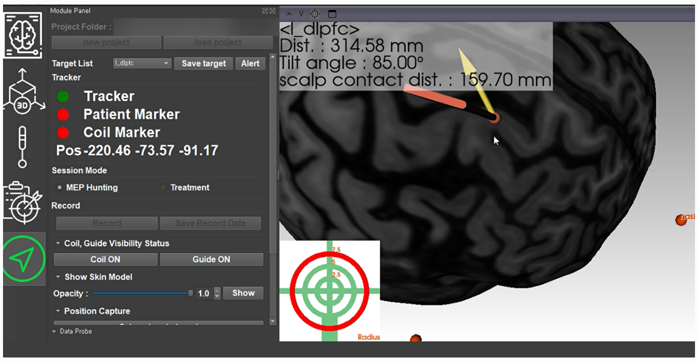
Verification and evidence ledger

03 / DEVICE AND SYSTEM INTEGRATION

Device and system integration

A major optical-tracker SDK migration, an in-house TCP binary protocol for the tablet, single tracker ownership, and contact-stop robot descent.

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

IMAGE / PUBLICLY INSPECTABLE

rtms-navigation-lead-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

rtms-navigation-gallery-01

Approved derivative; caption in portfolio data.

VERIFICATION AND EVIDENCE LEDGER

The evidence is the license-gated deployment build, a regression-test suite grown issue by issue, and coordinate-transform and registration logs.

Role, team result, and limits

04 / CLINICAL AND PRODUCT BOUNDARY

Clinical and product boundary

No clinical efficacy, quantitative accuracy, or regulatory status is claimed.

MY ROLE

Led the workflow, registration engine, device integration, and product structure.

Led the 3D Slicer procedure-setup and navigation workflow, the registration engine combining landmarks and ICP with its coordinate-transform chain, the optical-tracker SDK migration with the tablet protocol and

TEAM RESULT

AT&C owns license operations and adoption. I designed the core structure; a colleague built specific screens.

The evidence is the license-gated deployment build, a regression-test suite grown issue by issue, and coordinate-transform and registration logs.

LIMITATIONS

No clinical efficacy, quantitative accuracy, or regulatory status is claimed, and patient data stays out of this case.

NeuroPilot navigation module: tracker, patient-marker and coil-marker status, distance and tilt guidance to the target, and the aiming guide (project path hidden).

COLLABORATION BOUNDARY

The segmentation and landmark models were supplied externally and integrated behind an adapter boundary. I own that boundary, the post-processing, the license gating, and the UI integration.

Designed the 3D Slicer workflow from procedure setup to real-time coil navigation, the landmark-and-ICP registration engine, tracker, tablet, and robot integration, and a license-gated product structure.

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Recap and joint-development invitation

Turn optical tracking and 3D Slicer into a license-gated coil-navigation product that gives operators position and angle guidance they can rely on.

PRIMARY CAPABILITY

Medical Navigation & Visualization / 3D Geometry & Registration

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

The segmentation and landmark models were supplied externally and integrated behind an adapter boundary. I own that boundary, the post-processing, the license gating, and the UI integration.

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